Intervals and absolute value equations

Polynomial-time solution under the regularity promise

Difficulty: extreme **Importance:** broadly interesting

Status: Partially resolved

Area: algorithms for piecewise linear systems and complementarity

Rating rationale: Extreme reflects the longstanding polynomial-time barrier for P-matrix complementarity in an equivalent regular AVE form; broad impact spans optimization, complexity, and piecewise linear systems.

Partial resolution — 12 September 2026

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Theorem H, Sections 2.1–2.5 proves exact deterministic polynomial bit complexity under the original regularity promise when the rational input matrix is lower Hessenberg, including zero superdiagonal entries and zero solution coordinates. Theorem F gives a second subclass with a lower-triangular leading subsystem whose diagonal entries have absolute value greater than one, and one feedback variable. The additional structural assumptions are essential: arbitrary dense regular matrices remain unresolved. The original binary-input model and mathematical target below are unchanged.

The exact optimized-handicap family formula, exponential sign-region example and Newton-cycle certificates are supporting results; none proves an unrestricted complexity lower bound. These partial arguments passed a separate independent Codex AI-agent review. Submission record, verified affiliation and exact checks. AI assistance is disclosed; no historical priority, external human peer review or formal verification is claimed. No Lean verification was performed.

Context and notation

Absolute values and vector inequalities are componentwise. All algorithmic questions use rational input in binary and the deterministic Turing model. Input length includes the matrix dimensions and the bit lengths of all rational coefficients.

For a real square matrix A, define the diagonal perturbation family

$$\mathcal{D}(A) = \{A - \operatorname{diag}(d) : d \in [-1,1]^n\}.$$

Call this family **regular** when every member is nonsingular. Only diagonal entries vary. This is precisely the entrywise interval matrix $[A - I_n, A + I_n]$.

Problem statement

Is there a deterministic algorithm and a polynomial p such that, for every $n \geq 1$, rational $A \in \mathbb{Q}^{n\times n}$ and $b \in \mathbb{Q}^n$ with regular $\mathcal{D}(A)$, the algorithm outputs the exact rational vector x satisfying

$$Ax - |x| = b$$

within $p(\ell)$ bit operations, where ℓ is the binary input length?

The promise guarantees existence and uniqueness. No certificate of regularity is supplied or required, and behavior on inputs violating the promise is unrestricted. Polynomial dependence on coefficient bit length is allowed; this does not ask for a strongly polynomial algorithm.

References

Milan Hladík, Hossein Moosaei, Fakhrodin Hashemi, Saeed Ketabchi, and Panos M. Pardalos, *An overview of absolute value equations: from theory to solution methods and challenges*, Computational Optimization and Applications **93** (2026), 435–488, §2.4.2, paragraph following conditions (15)–(16); §2.2 gives the complementarity connection.

This is the AVE formulation of the polynomial-time P-matrix linear complementarity problem; it is counted once. Michaela Borzechowski, John Fearnley, Spencer Gordon, Rahul Savani, Patrick Schnider, and Simon Weber, *Two Choices Are Enough for P-LCPs, USOs, and Colorful Tangents*, ICALP 2024, §1, explicitly retain that algorithmic question.

Earlier status evidence — 2026-09-08

The survey appeared online August 8, 2025. Borzechowski et al.'s latest arXiv version is v2, May 21, 2024. Searches for "P-matrix linear complementarity 2026 polynomial-time" and "P-LCP solved polynomial algorithm" found no solution. E.-Nagy and Végh's June 30, 2026 revision, abstract and §6, gives an algorithm whose complexity also depends on an optimized handicap number; it does not give the required polynomial bound in input length alone.

Audit update — 2026-09-10

Rechecked the 2025 survey, §2.4.2, and E.-Nagy–Végh's 2026 handicap reduction. The latter's complexity also depends on its handicap parameter, so it does not establish a bound polynomial solely in the displayed input length. P-LCP/regular-AVE polynomial-time searches found no such algorithm or matching resolution.